

Jammming Project Initial Design Document

Muzaffer Karakuş

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1. Functionalities

Think about your apps general functionalities and write them down below. Write down the http request that will be required as a sub element to each functionality. Research through required login steps, and all the http request that will be required. Link the related documents beside each http request. When you update your whole app(When adding new functionalities, removing or updating functionalities), add new functionalities and related http requests at the bottom of the list following the same format.

- Login
 - Token Request
- Searching for a track
 - Search Request
- Displaying search results
- Adding a track to the playlist
 - Add a Track Request
- Removing a track from the playlist
 - Remove a Track Request
- Displaying your playlists
 - Playlists Request
- Creating a playlist
 - Create a Playlist Request
- Renaming playlist
 - Rename a Playlist Request
- Displaying playlist tracks
 - Playlist Elements Request

2. Function Components

After the previous step, it will be more obvious for you how your page is going to look. Next thing is to think about the information that you can display in separate divisions even if they are related. Starting from the App component, up to the grand children, write down the components that you will create. In detailed design of your components, you will update this part to include their children and their children. In addition, when you update your whole app(When adding new functionalities, removing or updating functionalities), you can update this part as required.

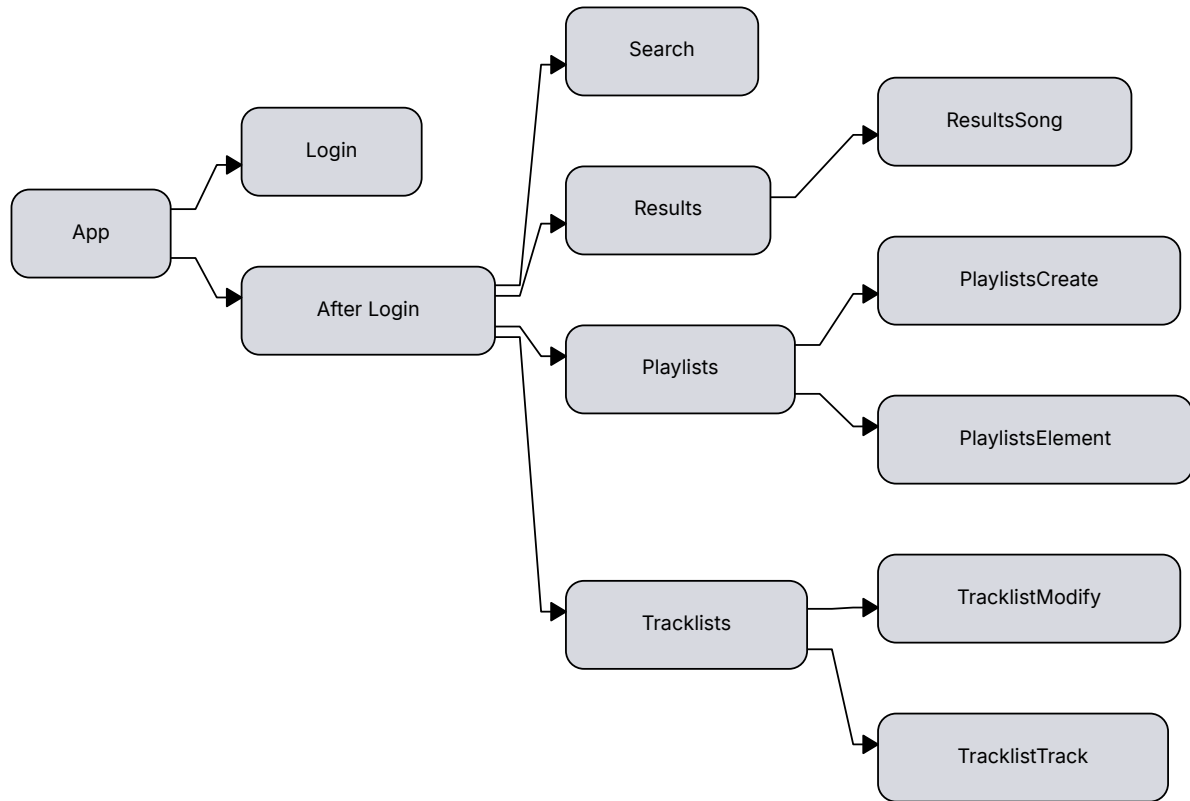


Figure 1: Component Chart

3. Events

Think about all the events that each component will include. Write them down for each component in separate tables. In separate columns, write down events with identifiable words, event type, event handler names you selected.

Event	Event Type	Event Handler
Sign in	click	handleSignInClick

Table 1: Events in **Login Component and its Children**

Event	Event Type	Event Handler
Search	click	handleSearchClick

Table 2: Events in **Search Component and its Children**

Event	Event Type	Event Handler
Add To Playlist	click	handleAddToPlaylistClick
Remove From Playlist	click	handleRemoveFromPlaylistClick

Table 3: Events in **Results Component and its Children**

Event	Event Type	Event Handler
Playlist	click	handlePlaylistClick
Create Playlist	click	handleCreatePlaylistClick
Cancel Playlist	click	handleCancelPlaylistClick
Save Playlist	click	handleSavePlaylistClick
Rename Playlist	click	handleRenamePlaylistClick
Cancel Rename	click	handleCancelRenameClick
Apply Rename	click	handleApplyRenameClick

Table 4: Events in **Playlist Component and its Children**

Event	Event Type	Event Handler
Remove From Tracklist	click	handleRemoveFromTracklistClick
Cancel Edits	click	handleCancelEditsClick
Save Edits	click	handleSaveEditsClick

Table 5: Events in **Tracklist Component and its Children**

4. Detailed Design

Do the detailed design for each component. Detailed design is done on child and grand child components of the App component.

During the design, first, functionalities and fetching operations related to this component are first rewritten.

Then the jsx that will be returned is formed. If there is a repetitive part or a part that you find much proper to separate, for each of these parts, a new child function component is designed. Everything these components will include in terms of jsx are specified here, not in a separate detailed design section for each child.

After these steps, coding is started, and respective coding operations are executed. During coding, states, effects, and helper functions that will be required are designed (Generally at the beginning of the coding). These must be documented to keep the track of them. This documentation is done on the last page of detailed design, including all of the components in one place. In this page, there will be 3 column grid.

The first column of the grid is for states. In this column there will be 2 column table, 1 column will be for state name, other column will be for where this state is used. In the second column of this table, you need to make the greatest parent bold to place the state in a proper parent.

The second column of the grid is for helper functions. In this column, each component, will have its name as level 3 heading. Under this heading, helper functions will be stated. When stating helper functions, it is necessary to create a different *helperFunction.js* for each current element or its children and their children. Below each file, functions exported should be stated.

The third column of grid is for effect hooks used. Similar to helper functions section, each component's name is used to create level 3 headings. Under these headings, in numbered lists, each effect hook is presented. Each list item includes bullet list with 3 elements. Rows in order are, dependency array, states and state setters and helper functions. All these elements are written when they are used inside these effects.

4.1. Login

4.1.1. Functionalities

- Sign in click
 - Request token

4.1.2. JSX



Figure 2: Login jsx

4.2. After Login

4.2.1. Functionalities

No functionality

4.2.2. JSX

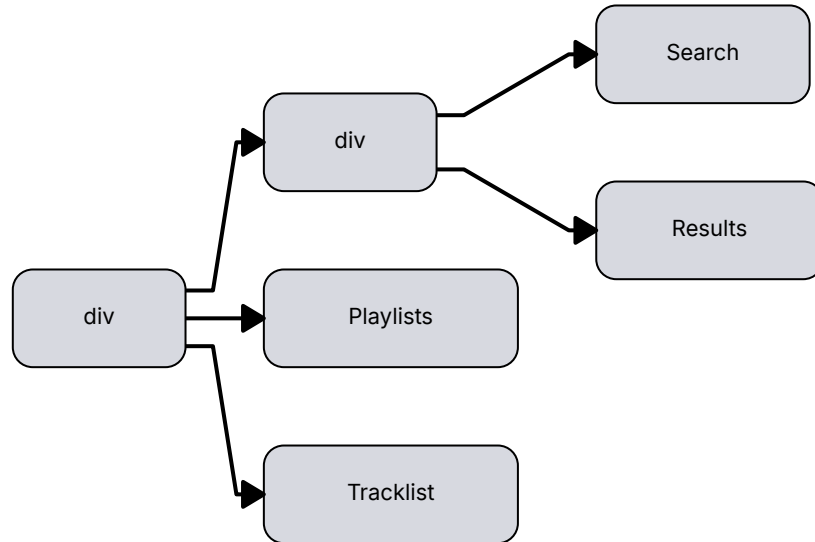


Figure 3: AfterLogin jsx

4.3. Search

4.3.1. Functionalities

- Search Click
 - Request search results

4.3.2. JSX

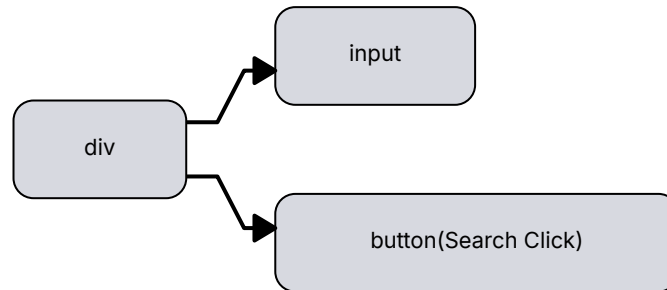


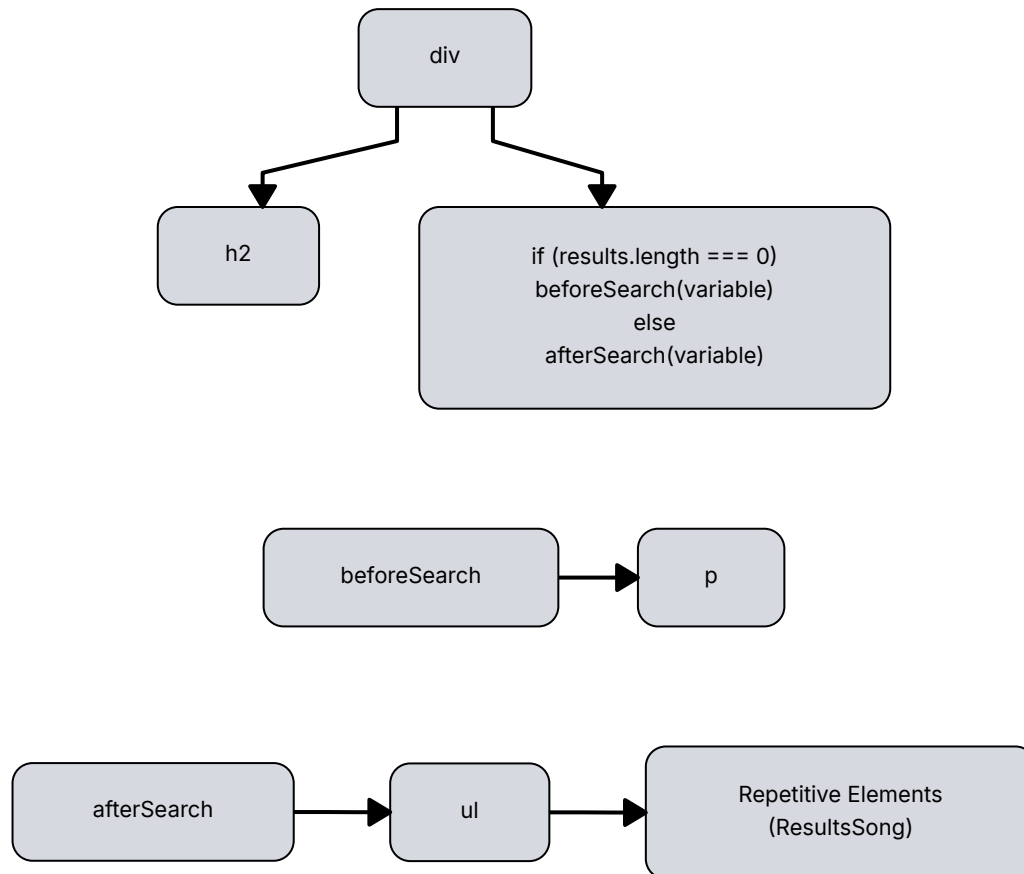
Figure 4: Search jsx

4.4. Results

4.4.1. Functionalities

- Add song click
- Remove song click

4.4.2. JSX



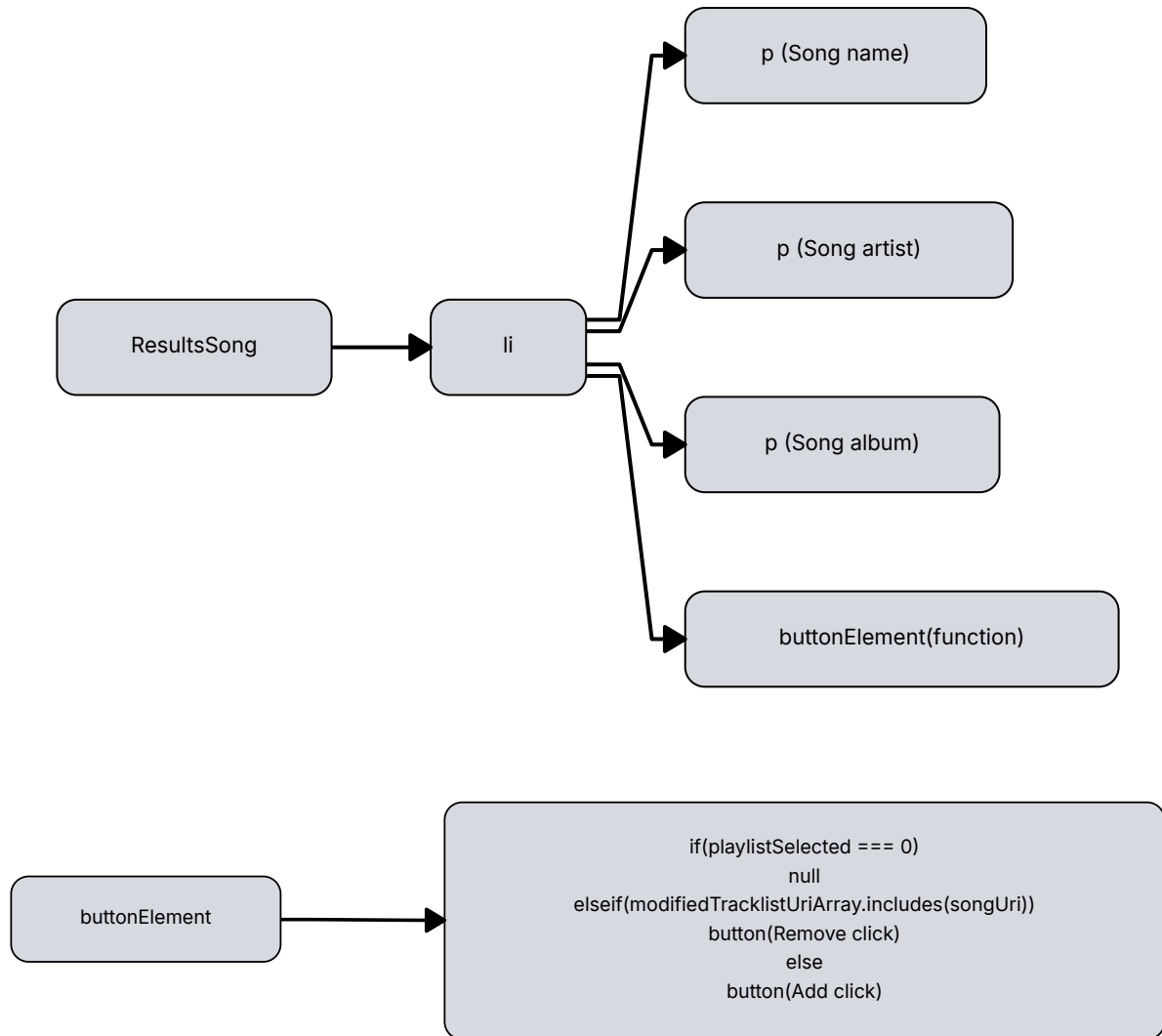


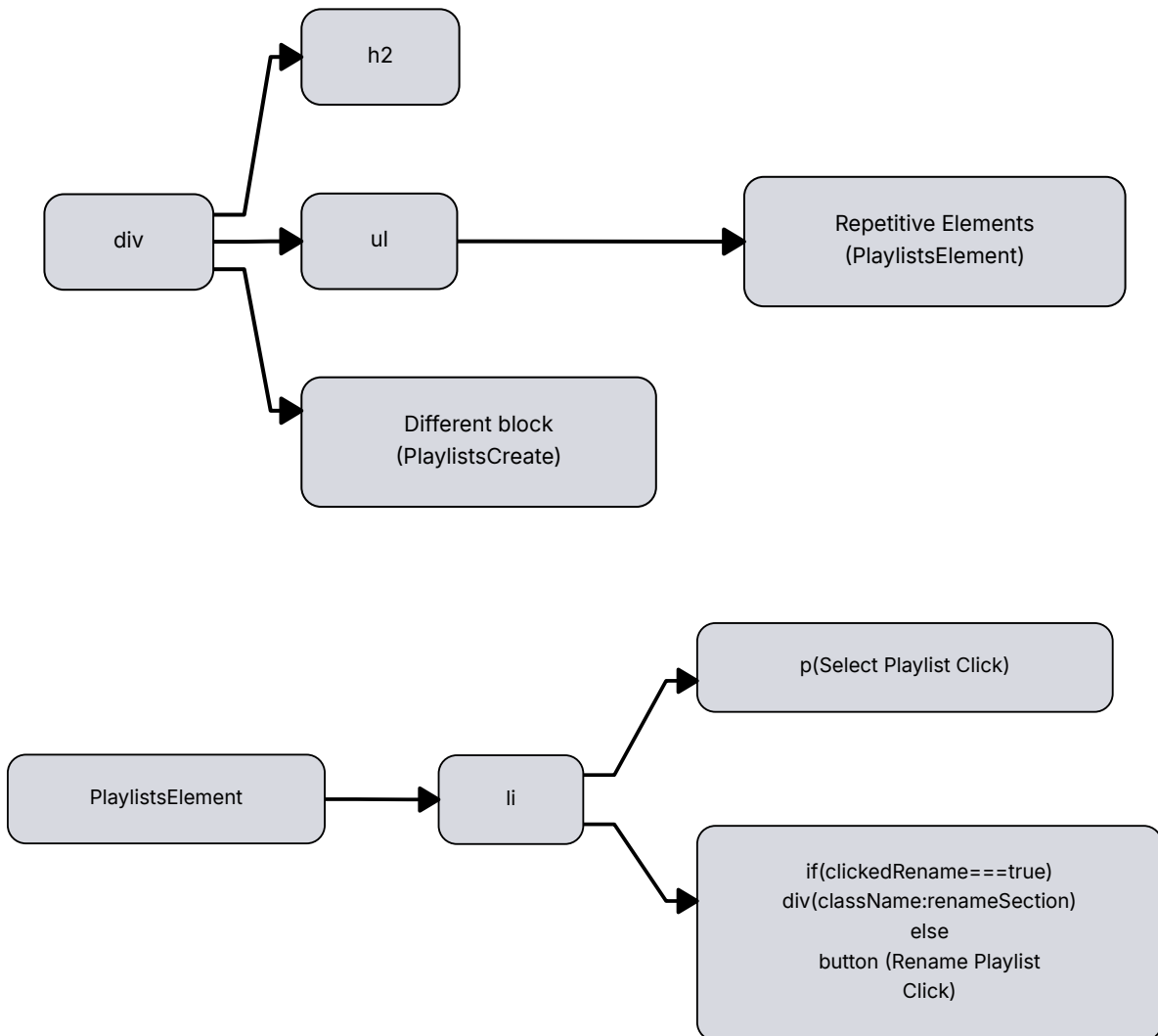
Figure 5: Results jsx

4.5. Playlists

4.5.1. Functionalities

- Request Playlists
- Rename Playlist Click
 - Apply Click
 - Request Playlist Rename
 - Cancel Click
- Create Playlist Click
 - Apply Click
 - Request Playlists
 - Cancel Click
- Select Playlist Click

4.5.2. JSX



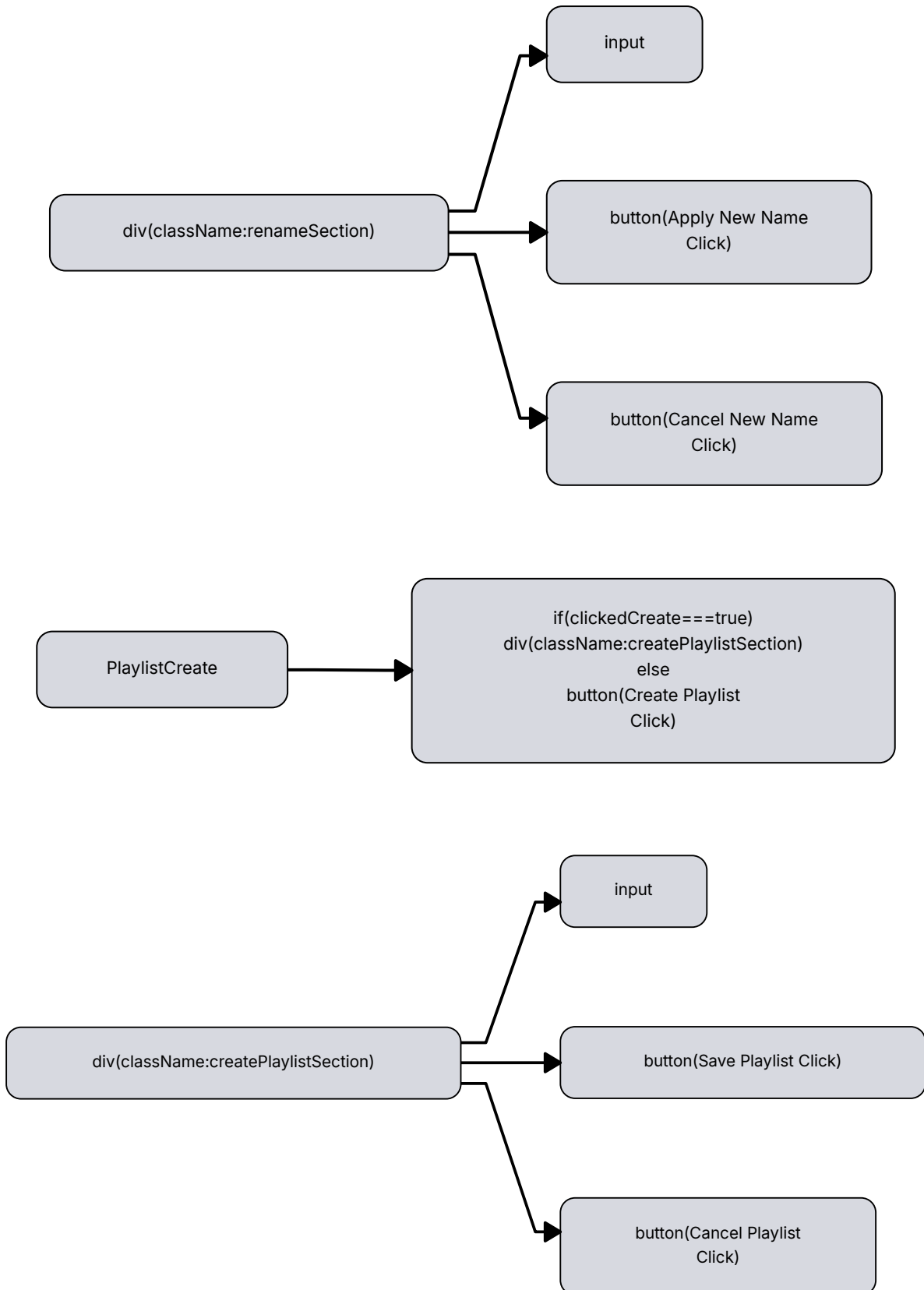


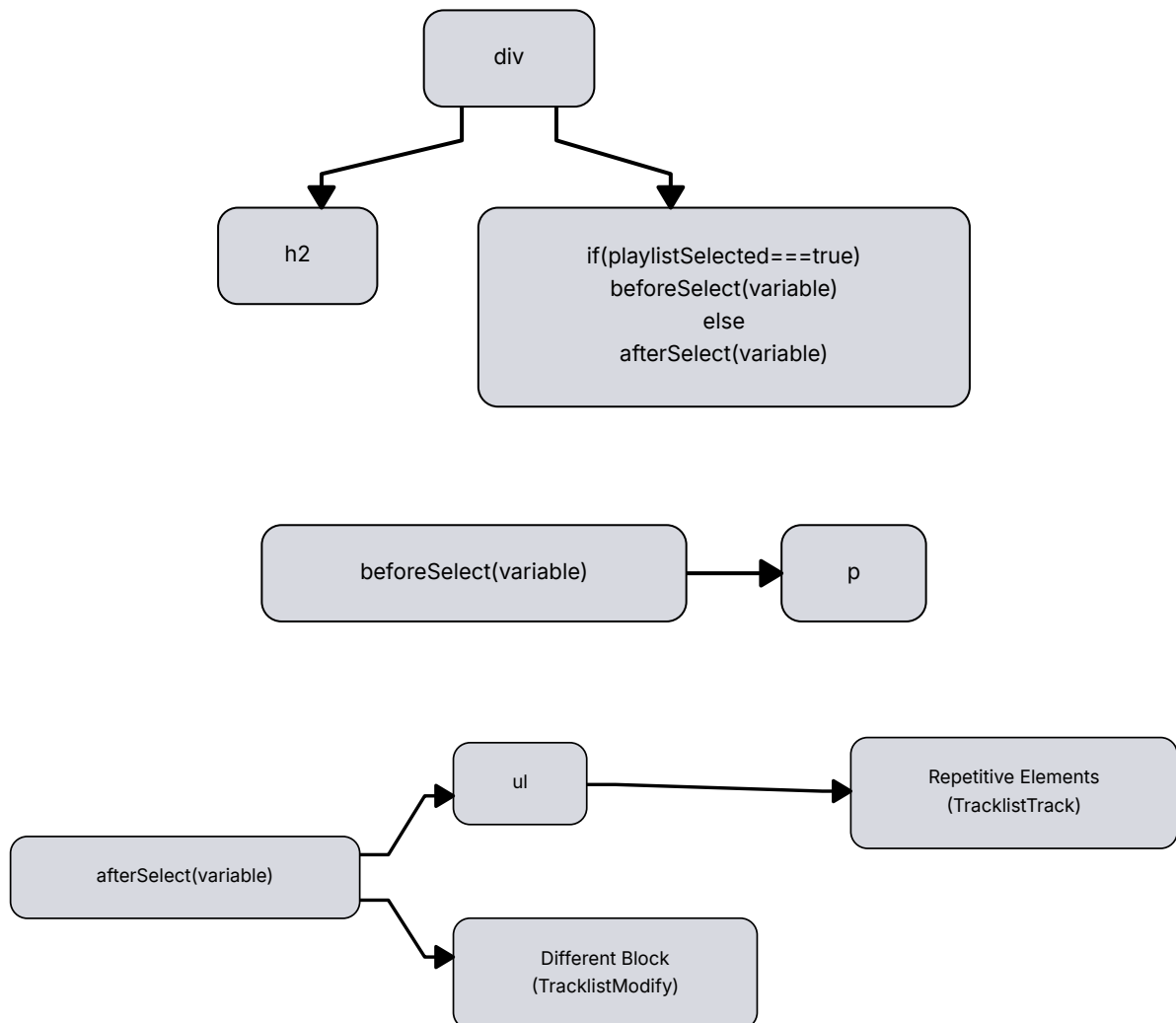
Figure 6: Playlists jsx

4.6. Tracklist

4.6.1. Functionalities

- Request Tracklist
- Remove song click
- Cancel tracklist update click
- Apply tracklist update click
 - Request add song to tracklist
 - Request remove song from tracklist
 - Request Tracklist

4.6.2. JSX



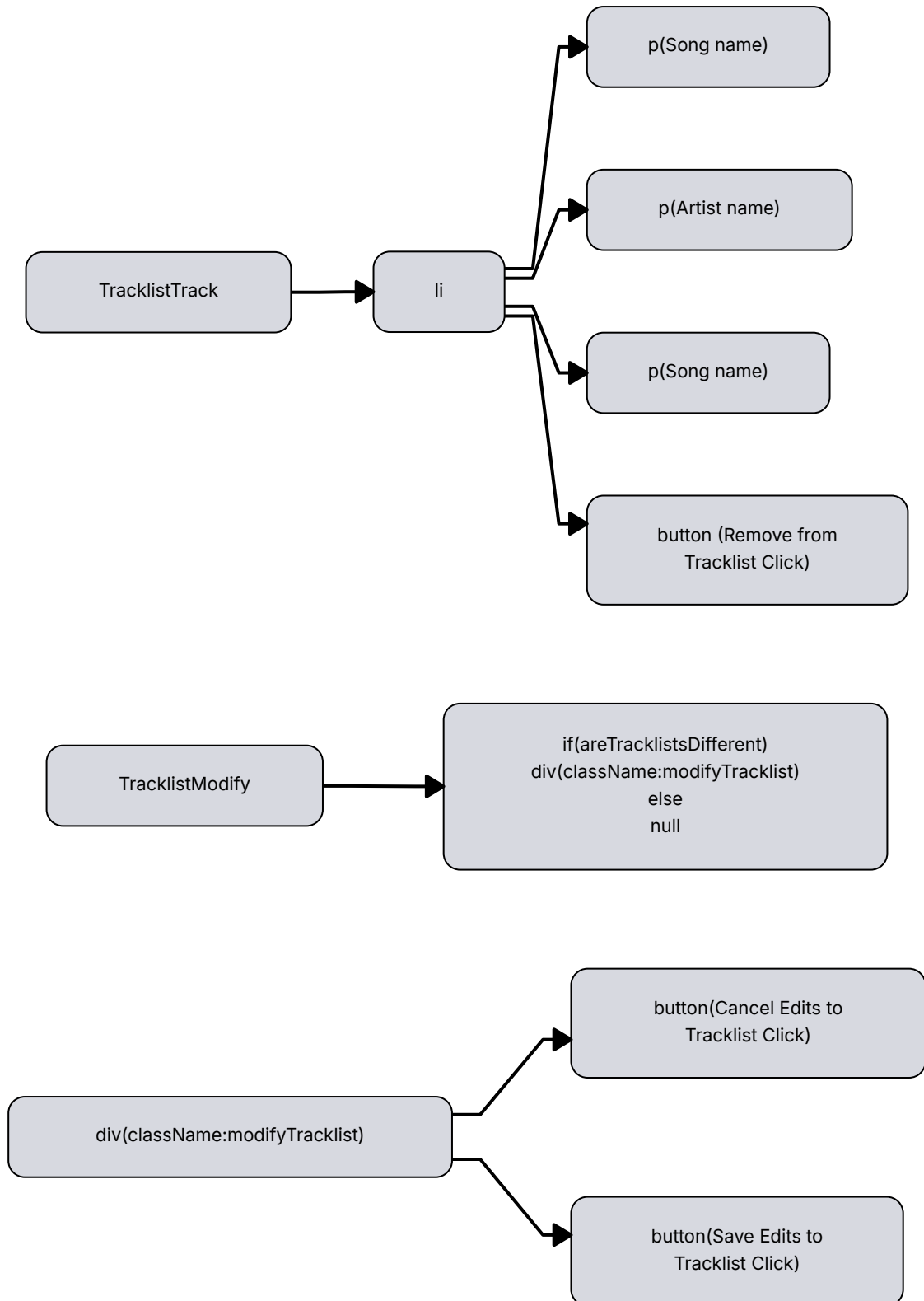


Figure 7: Tracklist jsx

4.7. States

State Name	Function Components
token	Login, App , AfterLogin,Search, Playlists
signedInjamming	Login, App
searchString	Search
results	Search, Results, AfterLogin
modifiedTracklist	Results, AfterLogin
playlistSelected	Results, AfterLogin
clickedRename	PlaylistsElement
newName	PlaylistsElement
clickNumberTo... TriggerEffect	PlaylistsElement, Playlists , PlaylistsCreate
newPlaylistName	PlaylistsCreate
clickedCreate	PlaylistsCreate
selectedPlaylist	AfterLogin , PlaylistsElement, Tracklist
originalTracklist	Tracklist
originalTracklist... UpdateCount	Tracklist , TracklistModify UpdateCount

Table 6: States table

4.8. Helper Functions

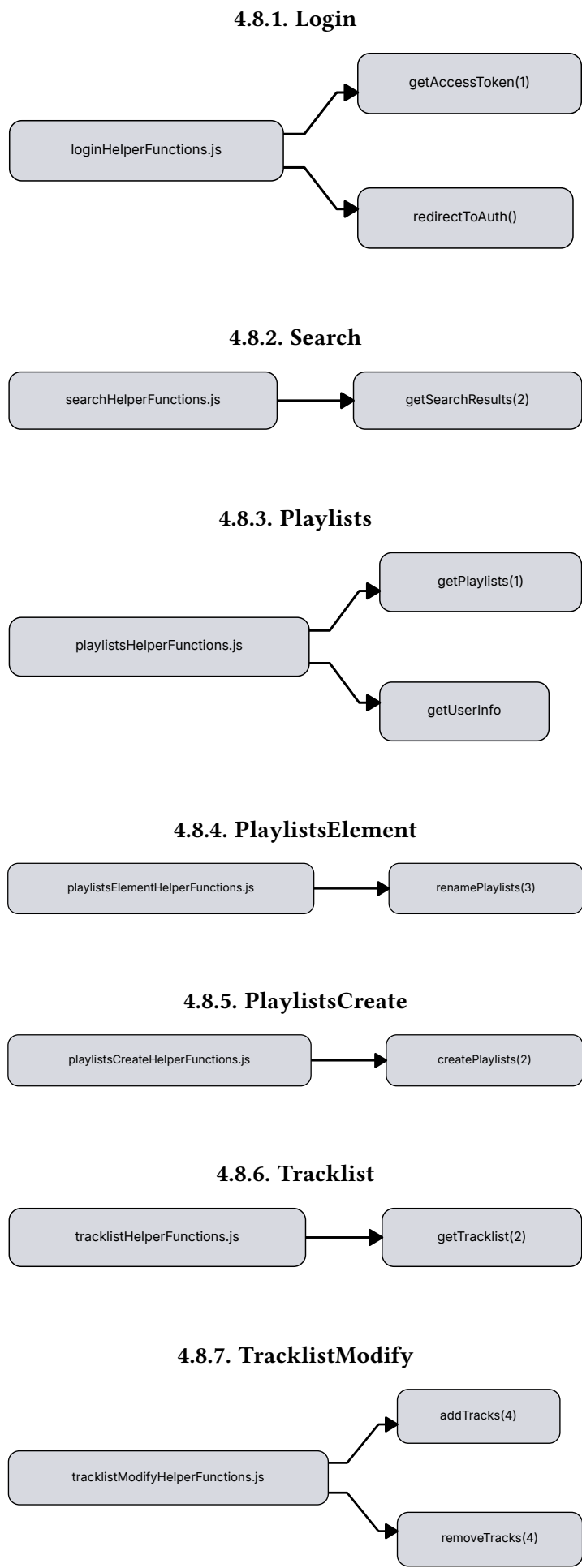


Figure 8: Helper functions

4.9. Effects Section

4.9.1. Login

1.
 - []
 - setToken, setSignedInJamming
 - getAccessToken from (loginHelperFunctions.js),

4.9.2. Playlists

1.
 - [clickNumberToTriggerEffect]
 - setPlaylists
 - getPlaylists, getUserInfo from (playlistHelperFunctions.js)

4.9.3. Tracklist

1.
 - [selectedPlaylists, originalTracklistUpdateCount]
 - setOriginalTracklist, setModifiedTracklist
 - getTracklist from (tracklistHelperFunctions.js)

5. Seperation of Presentational and Container Components

Each react function component should be seperated as presentational and container component.

6. Design of CSS

First, Blueprint for CSS should be detailly designed. Here, the grid elements, and where each button should be placed is determined.

Next, first general css file is created, second the modular css files for each component are determined.

Jamming

Search barSearch button Title: Search Results Results element Conditional Render See Figure 9 .	Title: Playlists Playlist NameRename playlist element Conditional Render See Figure 10. Playlist NameRename playlist element Conditional Render See Figure 10. Playlist NameRename playlist element Conditional Render See Figure 10. Playlist NameRename playlist element Conditional Render See Figure 10. Create playlist element Conditional Render See Figure 11.	Title: Tracklist Tracklist main element Conditional Render See Figure 12. Modify tracklist element Conditional Render See Figure 13.
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At first, below element is shown.

No search results found

If any search is done, below element below is shown.

Song name

Artist Name

Album Name

Add or
Remove button

Song name

Artist Name

Album Name

Add or
Remove button

Song name

Artist Name

Album Name

Add or
Remove button

Song name

Artist Name

Album Name

Add or
Remove button

Song name

Artist Name

Album Name

Add or
Remove button

Figure 9: Results Element

At first, below elements is shown.

Rename playlist button

If clicked on this element, element below is shown.

New Name Input

Cancel button

Apply button

Figure 10: Rename Playlist Element

At first, below elements is shown.

Create a new playlist button

If clicked on this element, element below is shown.

New Name Input

Cancel button

Apply button

Figure 11: Create Playlist Element

At first, below element is shown.

No playlist selected

If any playlist is selected, element below is shown.

Song name

Artist Name

Album Name

Remove button

Song name

Artist Name

Album Name

Remove button

Song name

Artist Name

Album Name

Remove button

Song name

Artist Name

Album Name

Remove button

Figure 12: Tracklist main element

At first, nothing is shown.

If any song is added or removed from the tracklist, element below is shown.

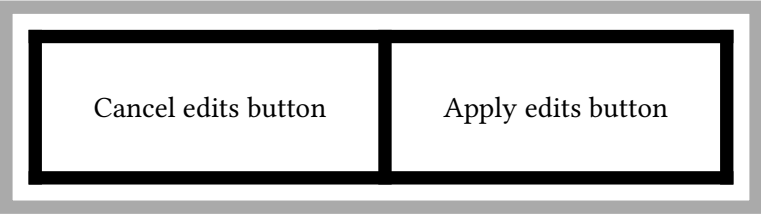


Figure 13: Modify Tracklist Element